Software Architecture and Process Models
INTE 22293- Academic Year 2024/2025
Session 1 : Introduction to the Module

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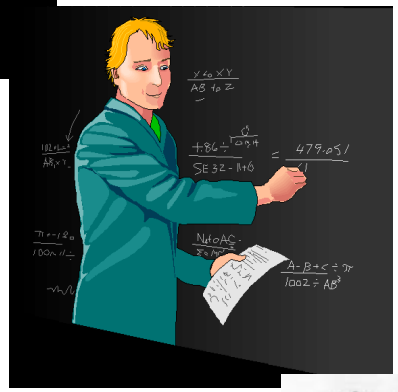
At Lectures ….

- **Look** - at what is being shown
- **Listen** - to what is being said
- **Think** - about the examples
- **Ask** - questions when you do not understand
- **Write** - when you think it is important



Teaching Methods…





Traditional





Innovative



Session Overview

- Introduction to the Module
- Discuss evaluation criteria
- Understand the challenges in Software Engineering
- Recognize the need of software architecture and process model
- Identify software requirements of an organization
- Digital Innovation and related terms
  - Digital transformation
  - Digitization and digitalization
  - Invention and innovation



Course Aim/Intended Learning Outcomes

- On completion of this course, the student should be able to:
  - Describe software development process models
  - Select suitable software development process model, and tools for software development
  - Apply system thinking and design thinking
  - Describe the importance of software architecture
  - Identify essential elements in software architecture
  - Describe different architecture Patterns
  - Describe quality attributes and basic rules of software design
  - Evaluate the effect of design on software quality
  - Design a software architecture to satisfy the system requirement



Course Content

- Software process models,
- System thinking and design thinking: generation of new ideas, problem solving and decision making,
- Software architecture and its importance, role of the architect, architectural drivers,
- Software architectural styles,
- Design suitable software architecture for a given system, architecture vs design,
- The factors that affect designs, the effect of design on software quality,
- Quality attributes of software design,
- Basic rules of software design,
- Design process: the input and start point of designs, the outcome and results of designs, transformation of data, satisfying and discovering constraints



Recommended Reading

- Humberto Cervantes, Rick Kazman (2016), Designing Software Architectures: A Practical Approach (SEI Series in Software Engineering) 1st Edition, Addison-Wesley Professional
- Mark Richards, Neal Ford (2020),Fundamentals of Software Architecture: An Engineering Approach 1st Edition by O'Reilly Media
- Zhu, H. (2005), Software Design Methodology: From Principles to Architectural Styles, Butterworth-Heinemann.
- Braude, E.J. (2003), Software Design: From Programming to Architecture, John Wiley.
- Sommerville, I. (2015), Software Engineering, 10th Edition, Pearson.
- Materials provided in the CAL



Evaluation Criteria

- Continuous assessment 20-40%
  - PBL
- Final examination 60-80%

- **At least 20% marks** from both components are required to pass the module.



Student Centered Learning

- You hear you forget
- You see you remember
- You do you understand



THREE DOMAINS OF LEARNING

Cognitive Domain
(Knowing/Head)

- Mental Skills *(KNOWLEDGE)*

Psychomotor Domain (Doing/Hand)

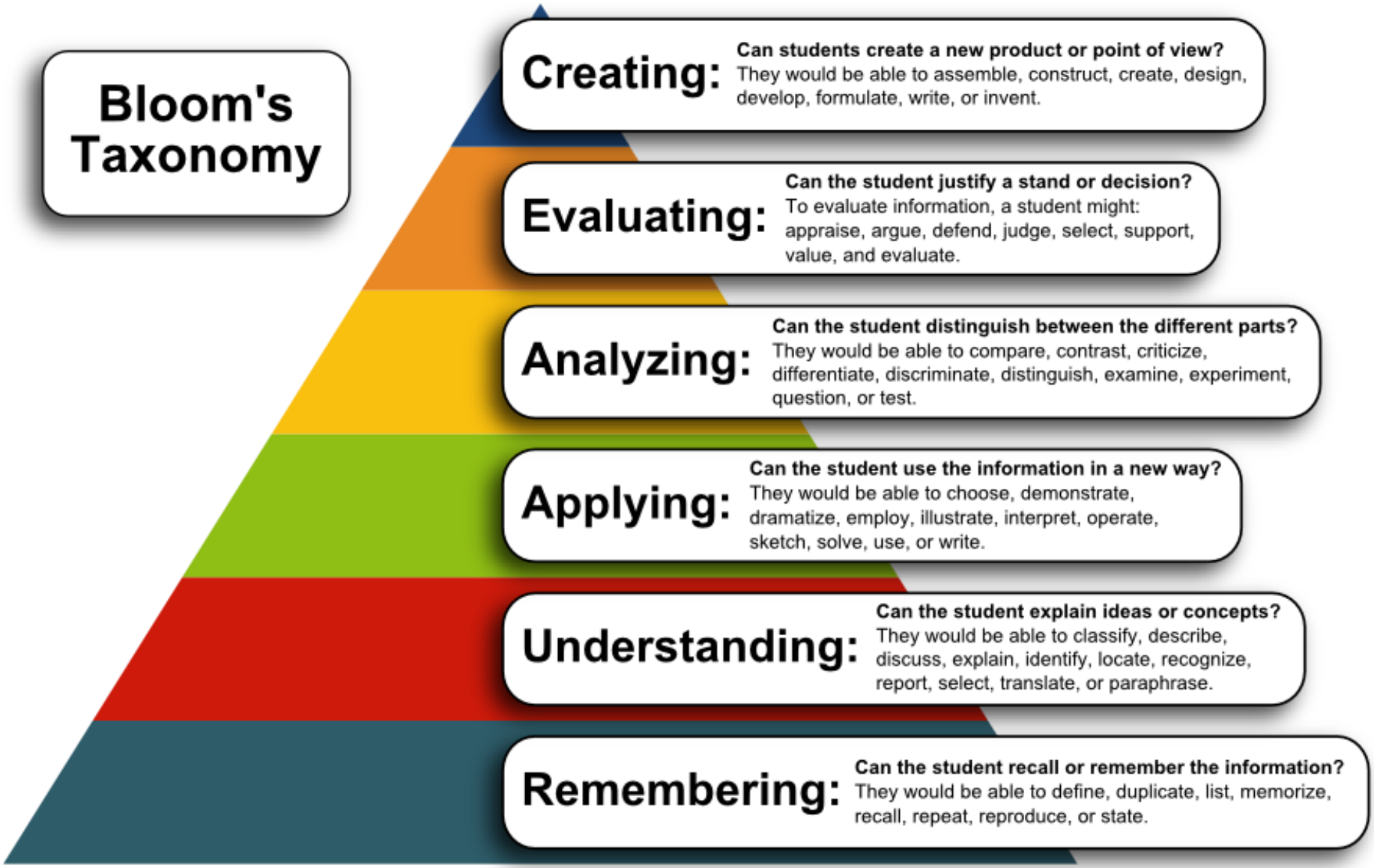
- Manual or physical skills *(SKILLS)*

Affective Domain
(Feeling/Heart)

- Growth in feelings or emotional areas *(ATTITUDE)*



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Bloom's Taxonomy
Creating:
Can students create a new product or point of view?
They would be able to assemble, construct, create, design, develop, formulate, write, or invent.
Evaluating:
Can the student justify a stand or decision?
To evaluate information, a student might: appraise, argue, defend, judge, select, support, value, and evaluate.
Analyzing:
Can the student distinguish between the different parts?
They would be able to compare, contrast, criticize, differentiate, discriminate, distinguish, examine, experiment, question, or test.
Applying:
Can the student use the information in a new way?
They would be able to choose, demonstrate, dramatize, employ, illustrate, interpret, operate, sketch, solve, use, or write.
Understanding:
Can the student explain ideas or concepts?
They would be able to classify, describe, discuss, explain, identify, locate, recognize, report, select, translate, or paraphrase.
Remembering:
Can the student recall or remember the information?
They would be able to define, duplicate, list, memorize, recall, repeat, reproduce, or state.

Learning Pyramid



Some Useful Terms….

- Literacy, IT Literacy, Data Literacy, Digital Dexterity, AI Literacy
- Invention, Innovation
- Digitization, Digitalization, Digital Transformation, AI Transformation





Invention vs Innovation

Digital Transformation

THE FOUR PATHS OF DIGITAL CHANGE



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*= Innovation has more then one definition | ** = According to Gartner | *** = According to Tim Herglotz

Transformations…..

- Digitization transforms information.
- Digitalization transforms processes.
- Digital Transformation transforms organizations.
- AI Transformation transforms decision-making.
- Human-AI Transformation transforms how people create value.



In Cass Activity 1

- What is your understanding of software architecture?
- What is your understanding of a software process model?
- Who decides the software architecture and software process model for a given project?
- When are the software architecture and software process model decided during a project?



Software Architecture

- High-level structure/organization of a software system,
- It includes fundamental components, their interactions, and the principles guiding their design and evolution.
- It provides a blueprint for the project, addressing concerns such as performance, scalability, security, and maintainability etc.
- It serves as a critical framework for guiding the development, deployment, and ongoing management of the software.



Software Process

- A software process is a structured set of activities, methods, and practices used to develop and maintain software.
- It defines the sequence of steps involved in creating a software product, including requirements gathering, design, coding, testing, deployment, and maintenance.
- The software process provides a framework for managing and improving the quality, efficiency, and predictability of software development,
- Often incorporating best practices and methodologies to ensure that the final product meets the desired specifications and stakeholder needs.



Software Architecture VS Software Process

- **Software architecture** is about *what* the system is and *how* it is constructed

- **Software process** is about *how* the software is developed, managed, and delivered.

- The architecture defines the technical structure of the system, while the software process defines the steps to create and maintain that structure



Software Architecture VS Software Process

- Software architecture and software process are distinct but complementary aspects of software development:
- **Software Architecture**:
  - **Focus**: The structure and design of a software system.
  - **Scope**: It defines the high-level components, their relationships, and the guidelines for building and evolving the system. It focuses on how the system is organized and how its parts interact to achieve specific objectives.
  - **Concerned with**: The system's technical blueprint, addressing aspects like performance, scalability, security, and maintainability.
  - **Outcome**: A well-defined architecture that serves as a guide for developers and stakeholders, ensuring that the system meets its technical and business goals.
- **Software Process**:
  - **Focus**: The sequence of activities involved in developing software.
  - **Scope**: It defines the methodologies, practices, and procedures followed during the software development lifecycle, including phases like requirements gathering, design, coding, testing, and deployment.
  - **Concerned with**: The workflow and management of the development process, aiming for efficiency, quality, and timely delivery.
  - **Outcome**: A systematic approach to software development that ensures consistent, repeatable results and adherence to project timelines and goals.



End of the Session 1



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